



وَفْرَة جرينتك
WafraGreentech

What changed in the mockup

Every change made to the Wafra Farm App UI mockup at version 1.5.4, round by round, in the order the comments arrived — the recorded call of 22 August and the four rounds of comment on the deck that followed it.

The build number is held at 1.5.4 on instruction: one deck, one number, while the review cycle is in flight. The requirement set underneath it is at 1.7.

Romain Bigare

Wafra Farm App · UI mockup v1.5.4 · Build Specification v1.7

What v1.5.4 changed, and where each of it landed

Round 1 of three. The comments on this deck cut six more screens and changed the icons, the plot shapes and the flow diagram; see [Mockup_Changes_v154_round2.md](#). Where the two disagree — the four judgement calls below in particular — the later document wins. The build number stays at 1.5.4 across all three rounds on instruction.

The companion to [Meeting_Comments_v152.md](#), which is the extraction from the call. This is the record of the build: every row in that table, the file it landed in, and the four judgement calls that were mine rather than the review's.

the specification to **1.6** — the first time the spec number has moved since v1.2 was issued as a document. Four earlier rounds redrew screens; this one deleted two whole concepts and changed a third, and no amount of redrawing would have implemented it. Both live in `app/meta.js`.

The five decisions, and what each one cost

1. Tasks are gone; advice is the only unit of work

The deepest change, and the one everything else hangs off. There is no task record, no task list, no assignment and no completion form.

An advice is now in one of four states, and `sentAt` is the whole of the difference between the first two:

STATE	WHAT IT MEANS
open, not sent	the farmer has not decided
open, sent	out with the supervisor, waiting for him to confirm
done	somebody recorded what was actually done, on D7
deferred	ignored or put off; it comes back tomorrow

`groupedTasks()`, `createTask()`, `completeTask()`, `blockTask()`, `assignAllAdvice()`, the `tasks` fixture array, the Tasks tab and WF3.004's badge, `TASK_ICON`, and the `tasks` entitlement key.

`unsentAdvice()`, `can('advice.send')`.

`tools/syntax.sh` now fails the build if the word "task" appears in live code outside a comment, or if any module other than `advice.js` calls `sendAdvice()`. Task management came back once before, one convenience button at a time.

2. The workforce is gone, and the worker role with it

\$5.6 was a directory: worker records, per-person languages, SMS/WhatsApp/push delivery pipes, invitations bound to records, and an identity model joining accounts to people by mobile number. All of it is deleted.

What replaced it is one selector — `supervisorOf(farmId)` — and one rule: work goes to that person as a message with a link they tap when it is done. Nobody below them holds an account.

The **worker role** went with it, which was not asked for in those words but follows from Mark's "there's only going to be one user": with no queue to hold and nothing to mark done, a worker account had nothing in it. The capability matrix ships two roles. A15 redeems an invitation as the farm's supervisor.

fixtures, assignees(), deliveryFor(), identityIds(), attachAccount(), accountForNumber(), the QR sheets, and the worker rows of the capability matrix, the notification categories and the cache/offline tables.

E6 and E7 survive and are **not** tasks — they are field capture, a person standing in a plot recording what the satellite cannot see. They moved to `app/screens/field.js`.

3. Single farm is the product

`router.homeRoute()` decides where Home opens: B1 for an account with more than one farm, B2:<the farm> for everybody else. It is computed rather than stored, so adding a second farm puts the chooser back without anyone remembering to.

Multi-farm stays in the requirements and in the build; it is simply not on the path a single-farm owner walks. Inside a farm, the app-bar title is a picker for accounts that have somewhere to go.

4. B2 and B3 merged

B3 is deleted. B2 is the farm **and** every plot on it: map, one line saying whether anything is urgent, the advice count, then the plots — crops first, tree groups second, under the state filter and the colour key that used to live on B3.

Three blocks came off the same screen in the same round:

- **"Health today"** — plant health, water stress and nutrition read at farm level. Every crop has its own profile, so the farm-level figure averages things that cannot be averaged. The measures are untouched on the plot and the map.
- **Weather** — moved to More, as the new **F15**. Given a screen of its own it prints the whole forecast the plan pays for rather than three columns and a "+11 days".
- **Workforce and Farm diary** — deleted with task management.

5. Trees are tree groups, and they must be surveyed

A tree group is one record per species per farm: `kind: 'trees'`, a species, a tree count, and a `parcels` count saying how many separate pieces of ground its trees stand on. It has no crop cycle — citrus is citrus — and no hand-drawn boundary, because trees are counted individually from the imagery and the count is what the price is worked out from.

Al Kharj North's **twelve date-palm plots are now one group of 7,801 palms**, standing in three places. Al Kharj South has four crop plots and four tree groups, which is why the deck opens B2 on it: farm-1 demonstrates the fold perfectly and the merged screen not at all.

The geometry change is contained: a plot is a list of rings (`plot.patches`) rather than one ring, `plot.geometry` is still the largest of them, and only `ui/map.js` had to learn about it. A group's parcels are drawn with a dashed outline so three shapes read as one holding.

survey and a sentence saying why, not a disabled card to argue with. A12 keeps the answer and the quote; it is the price screen now rather than the question screen.

The reminder that was new

Mark's case: it is January, the tomatoes are off, something else is in, and the satellite cannot name it for about three weeks because it reads a canopy.

A plot carries `harvestDetected0n`. While it is set, the plot has no current crop; the row on B2 says so in red with "Tell us what you planted", and B4 opens on the same question above the imagery rather than below three screens of it. `declareCrop()` clears it and opens a cycle. `plot-23` is in that state in the fixtures.

Everything else, by file

WHAT	WHERE
Urgent-only attention counts	<code>home.js</code> <code>urgentCount()</code> , and the wording of <code>b1.needning</code> / <code>b2.needning</code>
Four tabs, not five	<code>core/capabilities.js</code> <code>tabsFor()</code>
Home opens on the farm for single-farm accounts	<code>core/router.js</code> <code>homeRoute()</code>
Plot rows: name, size, what is growing, urgent flag; no NDVI	<code>home.js</code> <code>plotRow()</code>
Crops and tree groups as two blocks	<code>home.js</code> <code>B2()</code>
Plot list filter — all / urgent / action needed / watch / good	<code>state.ui.plotFilter</code>
Advice filtered by action type, not by assignee	<code>advice.js</code> <code>STATE_FILTERS</code> , <code>state.ui.adviceStateFilter</code>
Send / Ignore before, Record / Take it back after	<code>advice.js</code> <code>adviceCard()</code>
Send everything waiting, and an auto-send switch	<code>advice.js</code> <code>sendAllBar()</code>
Tree group detail: count, parcels, no cycle	<code>plot.js</code> <code>cropHeader()</code>

WHAT	WHERE
Boundary editing and crop cycles refused on a tree group	<code>overlays.js</code> <code>PLOT_MENU</code>
Weather	<code>more.js</code> <code>F15()</code>
Buttons say where they land, for a printed deck	<code>b1.seewhat.dest</code> , <code>b2.advice.dest</code>
Realistic plot geometry	unchanged — the shapes were already jittered parcels; what changed is that a farm now has four of them rather than twelve

Four judgement calls

Each is a small change back if it was read wrongly.

1. **Which four tabs.** The review agreed Tasks goes and that Plots earns a slot, and left the final set to me. Merging B2 and B3 made a Plots tab a second door onto the screen the farmer is already standing on, so the four are **Home, Map, Advice, More** and Home is the plot list. 2. **The worker role is deleted, not just the workforce screens.** Follows from "there's only going to be one user", but it was not said in those words. 3. **B3 is deleted rather than kept as a thinner screen.** "Merge the two" and "make B3 the home screen" are the same instruction from two directions; a surviving B3 would have duplicated B2. 4. **The activity log (F11) stays under More.** The "Farm diary" row on B2 is gone as asked, but F11 is the append-only audit log of WF5.187/WF5.188 rather than the diary the farmer was being offered, and deleting it would have taken `auditlog.view` with it.

Still open

- **One tree group per species, or one for all trees.** Mark is checking with Neil. The build goes per-species: lemon trees get different advice from date palms.
- **Who pays for the farmer's licence,** and pre-registration by phone number against ADAFSA's 25,500-farm list. The existing large-government registration path already covers it and needs no change yet.

Checks

`npm run check` · `npm run smoke` · `npm run deck`

The smoke test walks 579 renders across both roles and now also asserts: every farm has a supervisor to send work to; only the owner may send; an advice card carries Send/Ignore or Record/Take-it-back and never "Mark as complete"; and A9 offers one route for a farm of trees and two for a farm of crops.

The deck is `docs/Wafra_Farm_App_Screens_v1.5.4.pptx` — 62 slides, generated from the running app, so the screenshots cannot drift from the build.

v1.5.4, second and third rounds of comment

The build number stays at **1.5.4** on instruction: one deck, one number, while the review cycle is still in flight. The specification has moved twice underneath it and is at **1.7** — the two answer different questions, which is the point of holding both.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
 - Round 2 and 3, the comments on the deck: this document
-

Round 2 — everywhere

No screen names another screen

The build had begun annotating buttons with where they land — “*Opens your advice list (D1)*”. That belongs on the deck, drawn as arrows, not in an app a farmer uses. Gone.

Lucide, vendored

The ~90 hand-written path strings looked hand-drawn because they were. They come from **Lucide** now, read out of `lucide-static` by `tools/build-icons.mjs` into `app/ui/icons.data.js`. **Not a CDN**, though one was offered: this file is opened from `file://` and photographed by two headless browsers, and a network round trip is a way for any of those to come back with empty boxes. `npm run icons` regenerates.

Every plot is a rectangle

The wobbling five- and six-sided parcels and the centre-pivot circles were making a satellite mockup look like a hand drawing. The farm grid is square too, so parcels are no longer all portrait. The drawing canvas opens on a rectangle.

Six screens deleted

the `FARM_SWITCH` sheet now, carrying Add a farm), **B7** and **B8** (both were the map rebuilt at plot scope), **B9** (replaced by **B13**, the tree group),

Round 3 — screen by screen

SCREEN	WHAT CHANGED
B2	The plot list loses its card: hairlines between rows and nothing else, at <code>--ink-200</code> so they are actually visible without the box round them. The size moves up beside the plot name. Only the chevron navigates — the row itself is not a target, so the crop control beside it can be pressed without opening the plot. “Harvested — tell us what you planted” is “ Set the new crop ”, one line.
B12 / A9	Nothing is drawn disabled. The route cards greyed out until the farm had a name, which reads as broken rather than as sequenced and leaves the farmer guessing which field unlocks it. The name is still required — pressing a card without one puts the cursor in the name field and says why — but the choice looks like a choice from the moment the screen opens.

SCREEN	WHAT CHANGED
B4	The map pulls back: the pad round a plot scales with the plot rather than sitting at a fixed 20 m, so the boundary is no longer hard against the frame. "What is growing here now" goes from five stacked paragraphs to two lines — the question, then the fact and the button — with the satellite-phenology explanation behind an ①.
A3	Mobile / Email pills , per the example: a filled segmented control at the top, with the form beneath it swapping. It was one form with a "Log in with email and password" link at the bottom, which made one route the screen and the other a footnote a farmer had to read past his own form to find.
A5	Unchanged — the pills were tried here and taken out again. Registering is not a choice between two ways to prove who you are: an account needs both the mobile (it is the account) and the email (it finds a licence bought elsewhere), so a switch over two things you are going to fill in anyway only adds a step.
A12	Retitled " Your survey ". "What we will look for" is one line, and so is its answer. The explanation drops from five paragraphs to four short rows, the update-frequency line is gone, and the warning box is gone — the combined-service note is a sentence, because it is good news about the price rather than a caution.
A10 / A10D	The instruction — thirty-eight words wrapped over three lines above a map the farmer is trying to look at — moves behind a " How to draw this " chip. It is not lost, it is one tap away, and a farmer who has drawn one boundary never has to read it again.

The flow diagram is a tree

Undone in round 4. The tree was built, printed, and taken back out at the next review in favour of a linear strip that follows the section the page is filed under. Kept here because it happened; see [Mockup_Changes_v154_round4.md](#) for why.

The strip beside each phone was a straight line, which meant registration had to be declared twice — once ending in a survey, once in a drawing — and the deck could only ever show one of them. A farmer looking at the log-in page could not see from the paper that creating an account, redeeming an invitation and resetting a password all start there.

FLOWS is a DAG now: a root and, for each screen, what it leads to. The deck lays it out left to right by depth, stacks siblings, and draws orthogonal elbows. That buys three things a line could not say — where the app **branches** (A3 into three), where it **converges** (both drawing routes finish on A11), and what is a dead end.

Five trees, the deepest eleven columns. Tiles are sized from that one so the deck keeps a single rhythm, and a tree that would run past the footer shrinks until it fits — with a build-time warning naming the flow if it still cannot.

Checks

`npm run check` · `npm run icons` · `npm run smoke` · `npm run deck`

478 renders across both roles, no console errors. The smoke test learned this round: the plot list must be lines rather than cards with the crop as a control; A9 offers no route until the type is answered and no route cards at all to a farm of trees; A12 offers no choices; and A10's drawing instruction must be one tap away in the guidance sheet rather than absent.

v1.5.4, fourth round of comment

Still **1.5.4**, on instruction. One deck, one number, while the review cycle is in flight; the specification underneath it has moved to **1.7**.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
- Rounds 2 and 3, the first comments on the deck: `Mockup_Changes_v154_round2.md`
- Round 4, this document

The flow strip is a line again

Round 3 turned it into a tree. Round 4 turns it back, and the reason is worth writing down: a tree answers *what can this screen lead to*, which is a question about the app. A reviewer paging through a deck is asking *what was the farmer doing when he got here*, which is a question about one journey — and a diagram that shows every branch at once answers neither well on a printed page.

So `FLOWS` is a list of linear paths again, and each one now declares the

```
{ section: 'My Farm', name: 'Looking after the farm itself', ids: ['B2', 'B12'] }
{ section: 'My Plot', name: 'From the farm to one plot, and what is growing on it',
  ids: ['B2', 'B4', 'B5', 'B6'] }
{ section: 'Trees', name: 'From the farm to a tree group, and down to one tree',
  ids: ['B2', 'B13', 'B10'] }
```

B2 is on three of them. Which one prints beside it is decided by the section the page is filed under, so the copy of B2 in **My Farm** shows the farm journey and the copy in **My Plot** shows the plot journey. The reviewer's context picks the line, which is what "make sure the linear flow matches the current area in focus" asked for.

Fourteen paths across eight sections. Registration is two of them — one ending in a survey, one in a drawing — which is the price of a line, and a fair one.

Markers for the small buttons

"Add markers for small buttons that redirect to other slides, and small buttons that have important features behind them (like the map buttons). The markers should be located OUTSIDE of the phone mockup, and not cover any content."

A printout cannot be tapped. On paper an icon button that opens a whole screen and one that does nothing much look identical, and the three controls on B4's map — the reading picker, the date picker, the way into the map tab — are the clearest case: three glyphs, 40 dp each, carrying a third of what that screen can do.

in `app/ui/components.js`:

```
mapTool('scan', t('b4.openmap', 'Open in the map'), false, openMap, { deckTo: 'C1' })
helpButton(body, { title }) // declares its own note
overflowAction(onclick, label, { deckNote: 'Rename, edit the boundary, remove the plot' })
```


`deckTo` is the screen it leads to; `deckNote` is what it opens when it stays on this screen. Both render as `data-` attributes and **nothing appears in the app** — this is the same instruction as round 2's *no screen names another screen*, honoured rather than worked around. The attributes are invisible on a phone and read only by the deck builder.

`tools/screendeck.mjs` reads them off the rendered page with their positions and draws:

- a **numbered green disc in the gutter** — the 0.40 inch of white between the phone's right edge and the filmstrip, empty on every page of the deck — at the height of the control it names, so the eye travels straight across;
- a **numbered key** under the strip: `Open in the map → C1 Map (page 34)` for anything that leads somewhere, with the page number so the reviewer can turn to it, and the declared sentence for anything that opens in place.

Nothing is drawn over the phone and nothing is covered. Discs that would collide are pushed apart; the numbers still run top to bottom. Eight plot rows with the same chevron collapse to one marker rather than eight, and a marker pointing at a screen the deck does not print is dropped rather than sending the reviewer looking for a page that is not there.

DECK_OMIT

B11 was to come out of the slideshow. It was doing that by being left out of `SCREEN_GROUPS`, which quietly took it out of the app's own contact sheet too — a deck decision leaking into the app's index. `app/screens/index.js` now exports `DECK_OMIT = [ 'B11' ]`: the index stays complete, and the deck skips what the deck skips, in one named place.

Screen by screen

SCREEN	WHAT CHANGED
Global	Status vocabulary. Watch → Monitor , Action → Planned , with the meanings rewritten to say what to do: <i>No action required currently · Watch for changes and reassess · Complete as regular activity · Take action today</i> . The keys are unchanged, so nothing downstream had to move.
Global	Both calendars, everywhere. A date prints Gregorian and Hijri together — <code>12 February 2026 - 24 Sha'ban 1447</code> — ordered by the F8 setting (Gregorian first, Hijri first, or Hijri only). List cells and range endpoints stay short and Gregorian: two full dates either side of a dash is a paragraph where the column wants a stamp. B5's season bar was re-laid out for the extra width — the countdown moved above the bar and the two dates get a line each.
Global	Counts say urgent and count only urgent. The attention line counted everything that was not Good, which on any real farm says <i>something needs attention</i> every day of the year. It counts the urgent, and it is absent when there are none rather than printing "nothing urgent".
Global	A tree group is counted, not measured. Its hectares are the ground its parcels happen to cover — not what it is priced on, not what advice is calculated per, not a number anybody quotes about an orchard. Tree rows carry <code>1,240 trees</code> ; the hectares are gone from B13's header, from the B2 plot line and from A11.
A1	Register and Log in are the primary pair; the tour is a quiet filled container underneath rather than a third equal button.
A3	The code button reads "Send code to mobile number" .
A4 · A4A–A4D	The tour is five pages in the deck , one per panel, and sits after A14 — registration reads unbroken, then the detour that is offered on A1 as its own run.
FORGOT	Mobile and Email are the same switch as A3; the email route sends a reset link rather than a code, because that is what an email reset is.

SCREEN	WHAT CHANGED
A9	The fork — survey the whole farm, or draw my own plots — is offered only for field crops . A farm of trees is surveyed, because the tree count sets the price. Nothing is drawn disabled; pressing a route without a farm name puts the cursor in the name field and says why.
A10 / A10D	The drawing instruction lives behind a "How to draw this" chip. A10D gained the plot-type step it was missing. <i>(Round 5 moved the chip to an ⓘ on the app bar.)</i>
A11	Rectangles. The survey drew wobbling five- and six-sided parcels here after every other surface had been squared off. Same generator, same square grid, same look as B2 and the map. Tree rows show the number of trees, and the watch panel row is "Number of trees" .
A12	Retitled "Your survey" , cut to four short rows, warning box gone.
A13	Annual pricing on the plan cards at a 15% discount , the permission sentence on the trial card, and the bullets rewritten.
B2	The farm name at the top left is the picker, and it says so on hover; the plot list is hairlines rather than cards; only the chevron navigates.
B4	"See AI suggestions" → "See Advices" .
B11	Out of the deck — see <code>DECK_OMIT</code> above. Still in the app and still in the harness index.
B12	No longer re-offers the survey. <i>(Round 5 removed B12: it is A9B now.)</i>
F5 / F6	Carry the same annual discount as A13, from one exported constant.

Checks

```
./tools/syntax.sh · npm run icons · npm run catalogue · npm run smoke · npm run deck
```

494 renders across both roles, no console errors, 50 tiles for 50 screens on the contact sheet. The deck is 59 slides: cover, contents, 8 section dividers and 49 screens, 43 of which sit on one of the 14 paths.

v1.5.4, fifth round of comment

The last round on this version. Still **1.5.4**, on instruction; the requirement set underneath it is at **1.7**.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
- Rounds 2 and 3: `Mockup_Changes_v154_round2.md`
- Round 4: `Mockup_Changes_v154_round4.md`
- Round 5, this document

Adding a farm is one flow, and it is an A flow

"B12 should really be an A screen, and we need to remove the name input. When the user adds a farm, either through the FIRST RUN or through the MY FARM screen, we should present A9, and then this new version of B12, that leads to A10 or A10D."

B12 was in the wrong place twice over. It was filed under **My Farm**, which made adding a farm look like something you do to a farm you already have; and it asked for the farm's name a second time, in a draft of its own, because it ran the fork without A9 in front of it. Two screens described the same two routes, and they had already drifted apart once.

So the flow is one flow now, from either end:

```
A9  name the farm · how you measure land · what is growing on it
    ↓ Continue
A9B survey the whole farm, or draw the plots yourself
    ↓
A10 (survey)   or   A10D (draw)
```

notices that qualify it — the ten-farm limit and the combined-plan offer, both of which appear only for an account that already has farms. B12 is gone from the registry, from the app map and from the deck.

Everything that adds a farm now opens A9: the registration walk, and **Add a farm** in the farm picker at the top left of My Farm. `startAddFarm()` is the one door. *Adding a plot* to a farm that already exists is a different thing and was being routed through the add-a-farm flow by accident; it has its own `startDrawPlot()` and goes straight to the canvas, because the farm is already named and its type already settled.

A9 has a Continue button

It had none, and the reason it had none is the reason it needed one: the two route cards *were* the action, so a farmer who had filled in a name, a unit and a crop type had nothing to press. With the fork moved to A9B, A9 ends the way every other form in the app ends.

Not disabled. A dimmed button does not say which answer is short — this one lands on whichever one is, and says why.

Screen by screen

SCREEN	WHAT CHANGED
A9	Asks only: the farm's name, the unit its land is measured in, and what is growing on it. Continue in the dock. The fork and the trees explanation have moved to A9B.
A9B	New — the old B12, minus its name field. Two route cards for a farm of field crops; for a farm with any trees on it, the reason there is only one way in and a single Draw my farm boundary button. Carries the ten-farm limit and the combined-plan note for an account that already has farms.
B12	Removed.
A10 / A10D	The "How to draw this" chip is gone. The guidance is an ⓘ on the app bar, beside the subtitle that names what the screen is for — the same sheet, one tap, and the panel under the map is back to being about the plot rather than about the instructions. <code>helpChip()</code> has gone with it: there is one shape of help button in the kit now, not two.
A12	No card round We monitor / Priced by . Two facts in two lines did not need a box, and the box was the tallest thing on a screen whose whole point this round was that it had stopped being tall.

The deck

times over, which is a key repeated rather than a key.

produced a numbered disc the key could not explain, and a key line that began with a dash. The deck builder drops those now rather than drawing them, so a marker either says something or does not exist. Two related fixes went in with it: the ⓘ on a drawing screen names the screen it explains rather than saying

no-wrap setting introduced in round 4 is what made some of them come out blank in PowerPoint.

the cover, the contents and the eight section title pages leave the loop before any marker code runs. Worth saying plainly: on the generated deck I could not find a marker on the section 8 title page, in the file or in its XML. The guarantee is now structural rather than incidental, and if one is still showing in your copy, the slide number will tell me where to look.

Checks

```
./tools/syntax.sh · npm run smoke · npm run catalogue · npm run deck · npm run changelog
```

494 renders across both roles, no console errors, 50 tiles for 50 screens. The smoke test learned this round: A9 offers no route cards and must carry a Continue that refuses to proceed on an empty form; A9B is where the fork lives, gives a farm of trees no cards and a way on, and never asks for the farm name a second time; and A10's guidance is reachable from the **app bar** rather than from the body.

This document is generated: `tools/changelog.mjs` typesets these round records into `docs/Wafra_Mockup_Changes_v1.5.4.pdf`.